Xiaoyue Mei

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RESEARCH INTERESTS

Medicine, point-of-care diagnostics, biomarker discovery, aging, and tissue engineering.

EDUCATION

Ongoing:

University of California, San Francisco, and University of California, Berkeley — **Berkeley, CA**

Ph.D. Bioengineering from UCSF-UCB Joint Program — ***August 2023 - Present***

Completed:

University of California, Berkeley — **Berkeley, CA**

B.A. Molecular & Cellular Biology, minor in Data Science — ***September 2019 – May 2023***

- GPA: 3.758/4.00
- Relevant Courses:
  - Biochemistry and bioengineering: Organic Chemistry & Chemical Structure and Reactivity (A+), Introductory Physics (A), Comparative Virology (A), Cellular Engineering (TA for), Genetics, Genomics, & Cell Biology (A), Principles of Biochemistry & Molecular Biology (A-), Molecular Therapeutics (In Progress)
  - Data Science: Concepts of Probability (A), Principles & Techniques of Data Science(In Progress), Linear Algebra and Differential Equations(A), Multivariable Calculus(A)
- Honors & Fellowship:
  - Summer Undergraduate Research Fellowship(SURF) Fellow (summer 2022, Berkeley)
  - Dean's List (2021)

PUBLICATION AND PREPRINTS

[1] **Old plasma dilution reduces human biological age: a clinical study.**
Daehwan Kim, Dobri D Kiprov, Connor Luellen , Michael Lieb, Chao Liu, Etsuko Watanabe, **Xiaoyue Mei**, Kaitlin Cassaleto, Joel Kramer, Michael J Conboy, Irina M Conboy.
Published at *Geroscience*, 2022.

RESEARCH EXPERIENCE

[1] Systemic Impacts Of Therapeutic Plasma Exchange (TPE)
[2] Epigenetic Biomarkers of Aging [3] Cancer Proteomics — **Berkeley, CA**
[4] Identification of Biorthogonal Metabolic Protein — ***June 2021 – Present***
Advisor: Prof. Irina Conboy, Prof. Michael Conboy

[1] TPE is plasma dilution that replaces a patient's plasma with saline and purified albumin. We found it to promote a global shift to a younger systemic proteome, which have reduced cellular senescence and lower DNA damage.

- Proposed and implemented clustering tool UMAP to visualize the distribution and distinction of high dimensional data of 508 proteins from patients (treatment group), young, mid-age, old, neurological disordered control groups.
- Identified 72 proteins with statistically significant difference between young and old cohorts and performed Heatmap depicting the correlation of expression of each protein before and after each TPE treatment round 0 to 5.
- Annotate 679 age-variable CpGs to the candidate biological age-relevant genes/proteins based on the Conboy lab screens

[2] Epigenetic Project (with confidential information)

- Implemented programs for visualizing and analyzing high dimension datasets (with more than 480,000 CpG sites) through multi-dimensional UMAP and t-SNE to efficiently patternize DNA methylation in different cohorts.
- Implemented models for relative biological age of 6 different diseases and healthy control based on selected CpG reporters.

[3] Cancer Proteomics (with confidential information)

- Performed PCR to confirm whether or not mice and their offspring are expected mutant type, Click Western blot to identify the specific-chemical-labeled proteins, Conventional Western blot to measure the concentration of S100A10 proteins in mice model, and separate proteins with SDS-PAGE.
- Managed mice in behavioral testing including hanging test and treadmill test to evaluate mice muscle strength.
- Implemented t-SNE and UMAP to visualize the clustering effects of proteomics data in cancer and healthy mice cohorts.

[4] Identification of Bio-orthogonal Metabolic Protein (with confidential information)

- Implemented algorithm detecting 3% falsely labelled amino acid by mass spectrometry from 26309 DNA sequence segments with inconsistent length and order
- Analyzed and proposed possible reasons why each of the 7 unique amino acid is mismatched or misrecognized

Research Coordinator in Early Detection Of Cancer **Cross-strait Tsinghua Research Institute, Xiamen, China**
Advisor: Dr. Shanqing Lan ***Aug 2020 – Dec 2020***

- Established an online platform to efficiently track cell survival rates and total count from petri dish; prepare and maintain equipment for laboratory, participate in data cleaning and data organization for raw clinical trial data to ensure the accuracy.
- Helped with data presentation and plots replication of 4 research report, and draft questions to discuss in the ongoing research.
- Summarized and drafted a systemic review of 14 major biotech businesses in China and their services/products for early diagnosis of cancers available on market, serving as a reference for financial sponsorship to relevant research laboratories.

TEACHING EXPERIENCE

BioE 114 Cellular Engineering **Berkeley, CA**
Role: reader ***Aug 2022 – Present***

- Helped setup course page on bCourses and helped draft syllabus.
- Led occasional office hours answering students' questions on lecture contents and relevant publications.
- Counted attendance. Helped grade 80 students' assignments and projects and managed students' email inquiries.

ACTIVITIES

Bristol Hospice Volunteer Coordinator **Berkeley, CA,** ***Jan 2022 – Present***

- Visited 2 elderly with dementia in retirement house on a weekly basis, giving accompany to them and report their health conditions, mood, etc, to their family members.
- Helped organizing events including orientation for new volunteers and sending cards to patients on holidays.

Step Down Unit Volunteer **Oakland, CA**, ***Dec 2021 – Present***

- Dedicated 150+ hours to aid nurses and support patients including (assisting patient transfer, responding to patient calls, communicating patients' needs to nurses, ensuring organized storage of medical supplies, passing on patient reports to relevant medical professionals, etc) at the step-down unit of at Alameda Highland hospital.
- Directed patients, enhancing navigation within the hospital. Initiated hospital map software idea for improved accessibility.
- Assisted security in patient check-in, enhancing admission procedures.
- Provided translation for 30+ patients, improving communication and care.

Commercial designer **Remotely , CA**, ***Oct 2019 – Present***

- Edited video and designed project website to promote and organize sale and other data about products from a Sonoma farm.
- Gained 11k total views for the product promotion videos, positive feedbacks about high responsibility when working remotely.

SKILLS

- **Medical Skill:** Certified EMT by Shenzhen Center For Prehospital Care, Proficient in using Automated External Defibrillator (AED) and skilled in conducting Basic Life Support (BLS) procedures.
- **Programming Languages (with descending proficiency):** Python, MATLAB, R, Java.
- **Software:** LaTeX, BioRender, Excel, Prism.
- **Skills:** PCR, Fluorescence Microscopy, 3D Printing, Western Blots, BioMEMS fabrication.